



Decimals & Percentages

A Focused Guide to Decimal Operations & Percent Skills



Target Audience: Grades 5–8

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"Dreams can come true. Better late than never."

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Section 1: Understanding Decimals

A **decimal** is a number that uses a decimal point to show values less than one whole. Decimals are based on the number 10 — each place to the right of the decimal point represents a fraction with a denominator that is a power of 10.

Place Value Chart:

Hundreds	Tens	Ones	.	Tenths	Hundredths	Thousandths
100	10	1	.	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$
100	10	1	.	0.1	0.01	0.001

Reading Decimals:

Read the whole number part, say "and" for the decimal point, then read the decimal digits using the place value of the **LAST** digit.

Comparing Decimals:

Line up the decimal points. Compare digit by digit from left to right. Add zeros to make the same number of decimal places if needed.

Worked Examples:

Example 1: Read 3.45 in words

"Three and forty-five hundredths"

(Last digit is in hundredths place)

Answer: Three and forty-five hundredths

Example 2: Which is greater: 0.7 or 0.65?

Add zero to make same length: 0.70 vs 0.65

Compare tenths: $7 > 6$

Answer: 0.7 is greater

Section 1 (continued)

Example 3: Write 0.025 as a fraction

$$0.025 = 25 \text{ thousandths} = 25/1000$$

$$\text{Simplify: GCF}(25, 1000) = 25$$

$$25 \div 25 = 1, 1000 \div 25 = 40$$

Answer: 1/40

Example 4: Round 4.678 to the nearest tenth

Look at the hundredths digit: 7

$7 \geq 5$, so round UP the tenths digit

$$4.6 \rightarrow 4.7$$

Answer: 4.7

Example 5: Order from least to greatest: 0.3, 0.28, 0.305

Make same length: 0.300, 0.280, 0.305

Compare: $280 < 300 < 305$

Answer: $0.28 < 0.3 < 0.305$

Key Decimal Facts:

- Adding a zero at the END of a decimal does NOT change its value: $0.5 = 0.50$
- Moving the decimal RIGHT makes the number LARGER ($\times 10$ each place)
- Moving the decimal LEFT makes the number SMALLER ($\div 10$ each place)

Section 2: Decimal Operations

Adding, Subtracting, and Multiplying

Adding & Subtracting Decimals:

Step 1: Write the numbers vertically, lining up the decimal points.

Step 2: Fill empty places with zeros.

Step 3: Add or subtract column by column (right to left).

Step 4: Bring the decimal straight down into your answer.

Worked Examples — Adding & Subtracting:

Example 1: $3.45 + 2.7$

Line up decimals and add zero:

```
3.45
+ 2.70
-----
```

6.15

Answer: 6.15

Example 2: $10.0 - 4.38$

Line up and add zeros:

```
10.00
- 4.38
-----
```

5.62

Answer: 5.62

Example 3: $5.6 + 0.045 + 12.3$

Line up:

```
5.600
```

$$\begin{array}{r} 0.045 \\ +12.300 \\ \hline \end{array}$$

$$17.945$$

Answer: 17.945

Example 4: $8.2 - 3.756$

Line up:

$$\begin{array}{r} 8.200 \\ - 3.756 \\ \hline \end{array}$$

$$4.444$$

Answer: 4.444

Multiplying Decimals:

Step 1: Ignore the decimal points — multiply as whole numbers.

Step 2: Count the TOTAL decimal places in BOTH factors.

Step 3: Place the decimal point that many places from the RIGHT in your answer.

Worked Examples — Multiplying:

Example 5: 2.5×0.4

Multiply as whole numbers: $25 \times 4 = 100$

Count decimal places: 2.5 has 1, 0.4 has 1 $\rightarrow$ total = 2

Place decimal 2 places from right: 1.00

Answer: 1.0

Example 6: 3.14×2.5

Multiply: $314 \times 25 = 7850$

Decimal places: 3.14 has 2, 2.5 has 1 $\rightarrow$ total = 3

Place decimal: 7.850

Answer: 7.85

Example 7: 0.06×0.3

Multiply: $6 \times 3 = 18$

Decimal places: 0.06 has 2, 0.3 has 1 $\rightarrow$ total = 3

Place decimal: 0.018

Answer: 0.018

Example 8: 12.4×0.05

Multiply: $124 \times 5 = 620$

Decimal places: 12.4 has 1, 0.05 has 2 $\rightarrow$ total = 3

Place decimal: 0.620

Answer: 0.62

Section 3: Converting Between Forms

Fractions $\leftrightarrow$ Decimals $\leftrightarrow$ Percentages

Fraction $\rightarrow$ Decimal: Divide the numerator by the denominator.

**Decimal $\rightarrow$ Percent: Multiply by 100 (move decimal 2 places RIGHT).
Add % sign.**

**Percent $\rightarrow$ Decimal: Divide by 100 (move decimal 2 places LEFT).
Remove % sign.**

Percent $\rightarrow$ Fraction: Write over 100, then simplify.

Worked Examples:

Example 1: Convert $\frac{3}{4}$ to a decimal and percent

Fraction $\rightarrow$ Decimal: $3 \div 4 = 0.75$

Decimal $\rightarrow$ Percent: $0.75 \times 100 = 75\%$

Answer: $0.75 = 75\%$

Example 2: Convert 0.6 to a fraction and percent

Decimal $\rightarrow$ Fraction: $0.6 = \frac{6}{10} = \frac{3}{5}$

Decimal $\rightarrow$ Percent: $0.6 \times 100 = 60\%$

Answer: $\frac{3}{5} = 60\%$

Example 3: Convert 45% to a decimal and fraction

Percent $\rightarrow$ Decimal: $45 \div 100 = 0.45$

Percent $\rightarrow$ Fraction: $\frac{45}{100} = \frac{9}{20}$

Answer: $0.45 = 9/20$

Section 3 (continued)

Example 4: Convert $\frac{1}{8}$ to a decimal and percent

Fraction $\rightarrow$ Decimal: $1 \div 8 = 0.125$

Decimal $\rightarrow$ Percent: $0.125 \times 100 = 12.5\%$

Answer: $0.125 = 12.5\%$

Example 5: Convert 150% to a decimal and fraction

Percent $\rightarrow$ Decimal: $150 \div 100 = 1.50$

Percent $\rightarrow$ Fraction: $150/100 = \frac{3}{2} = 1 \frac{1}{2}$

Answer: $1.5 = \frac{3}{2}$

Example 6: Convert $\frac{2}{3}$ to a decimal and percent

Fraction $\rightarrow$ Decimal: $2 \div 3 = 0.6666... \approx 0.667$

Decimal $\rightarrow$ Percent: $\approx 66.7\%$

Answer: $0.667 \approx 66.7\%$

Common Conversions Reference Table:

Fraction	Decimal	Percent
$\frac{1}{2}$	0.5	50%
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{1}{3}$	0.333...	33.3%
$\frac{1}{8}$	0.125	12.5%
$\frac{3}{8}$	0.375	37.5%

Section 4: Finding Percent of a Number

Formula: Percent $\times$ Whole = Part

To find a percent of a number:

Step 1: Convert the percent to a decimal (divide by 100).

Step 2: Multiply the decimal by the number.

Three Types of Percent Problems:

- **Find the PART:** What is 30% of 60? $\rightarrow 0.30 \times 60 = 18$
- **Find the WHOLE:** 15 is 25% of what? $\rightarrow 15 \div 0.25 = 60$
- **Find the PERCENT:** 12 is what % of 80? $\rightarrow 12 \div 80 = 0.15 = 15\%$

Worked Examples:

Example 1: Find 20% of 85

Convert: $20\% = 0.20$

Multiply: $0.20 \times 85 = 17$

Answer: 17

Example 2: Find 75% of 200

Convert: $75\% = 0.75$

Multiply: $0.75 \times 200 = 150$

Answer: 150

Example 3: Find 6% of 50

Convert: $6\% = 0.06$

Multiply: $0.06 \times 50 = 3$

Answer: 3

Example 4: A shirt costs \$40 and is 25% off. What is the discount?

Find 25% of \$40:

$$0.25 \times 40 = \$10 \text{ discount}$$

$$\text{Sale price: } \$40 - \$10 = \$30$$

Answer: \$10 discount (sale price = \$30)

Example 5: You scored 18 out of 20 on a quiz. What percent?

$$\text{Divide: } 18 \div 20 = 0.9$$

$$\text{Convert: } 0.9 \times 100 = 90\%$$

Answer: 90%

Section 5: Practice Drill

35 Problems — Decimals & Percentages

Solve each problem. Show your work.

1) $4.56 + 3.7 =$

2) $12.03 - 8.9 =$

3) $2.5 \times 3.2 =$

4) $6.4 \div 0.8 =$

5) Convert $\frac{5}{8}$ to a decimal.

6) Convert 0.35 to a fraction (simplify).

7) Convert 0.72 to a percent.

8) Convert 85% to a decimal.

9) Find 40% of 90.

Practice Drill (continued)

10) Find 15% of 200.

11) What percent is 9 of 36?

12) $0.003 \times 100 =$

13) $7.25 - 4.008 =$

14) Convert $\frac{3}{20}$ to a decimal and percent.

15) A book costs \$24. Tax is 8%. What is the total cost?

16) $0.45 + 0.6 + 1.2 =$

17) $5.04 \times 0.6 =$

18) Convert 125% to a decimal.

Practice Drill (continued)

19) 20 is what percent of 80?

20) Find 5% of 360.

21) Order from least to greatest: 0.5, 0.48, 0.503

22) Round 6.7849 to the nearest hundredth.

23) Convert $\frac{7}{9}$ to a decimal (round to thousandths).

24) $3.6 \times 0.25 =$

25) $15.6 \div 1.2 =$

26) What is 90% of 110?

27) Convert 0.008 to a percent.

Practice Drill (continued)

28) $8.05 - 2.997 =$

29) A student got 42 out of 50 correct. What percent?

30) Find 12.5% of 80.

31) $0.7 \times 0.7 =$

32) Convert $11/25$ to a percent.

33) $4.2 + 0.38 + 10 =$

34) 33 is what percent of 60?

35) Find 200% of 45.

Section 6: Answer Key

- 1) 8.26
- 2) 3.13
- 3) 8.0
- 4) 8
- 5) 0.625
- 6) $\frac{7}{20}$
- 7) 72%
- 8) 0.85
- 9) 36
- 10) 30
- 11) 25%
- 12) 0.3
- 13) 3.242
- 14) $0.15 = 15\%$
- 15) \$25.92
- 16) 2.25
- 17) 3.024
- 18) 1.25

Answer Key (continued)

- 19) 25%
- 20) 18
- 21) 0.48, 0.5, 0.503
- 22) 6.78
- 23) 0.778
- 24) 0.9
- 25) 13
- 26) 99
- 27) 0.8%
- 28) 5.053
- 29) 84%
- 30) 10
- 31) 0.49
- 32) 44%
- 33) 14.58
- 34) 55%
- 35) 90